

INVESTMENTS & PORTFOLIO MANAGEMENT PROJECT- 2

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SECTION 1- Data and Descriptive Analysis

1. The correlation value indicates a moderate positive relationship between the monthly returns of TGT and WMT

	TGT	WMT
Average	0.0126381	0.0122069
Sample Standard Deviation	0.0898585	0.0533014
Sharpe	0.118673	0.191494
Correlation	0.432	0.432

2. A monthly risk-free rate of **0.2%** was used in the calculation.

Weight TGT	Weight WMT	Portfolio return	Portfolio risk	Portfolio sharpe
0.218799	0.781201	0.0123012	0.0531761	0.193720

SECTION 2 - CAPM and Fama French Models

1. About Target

Target Corporation, founded in 1962 and headquartered in Minneapolis, is a premier American general merchandise retailer operating nearly 2,000 stores across all 50 U.S. states as of 2025.

It provides a wide range of products including apparel, beauty, electronics, home goods, and food through its physical locations and digital channels such as: Target.com. The retail industry is highly competitive and sensitive to economic cycles, which makes Target's stock performance responsive to overall market conditions. As a leading retailer, Target competes with major companies such as Walmart, Amazon and Costco Wholesale. In addition to its retail stores, Target operates in supply chain management and has a significant digital presence.

2. Excess Returns

To estimate the CAPM and Fama-French models, excess returns were calculated as:

$$\text{Excess Return (TGT)} = \text{TGT Return} - \text{Risk-Free Rate (RF)}$$

The Fama-French monthly risk-free rate (RF) was used.

The market factor used in both models was the Fama-French Market Excess Return (Mkt – RF).

These excess returns were then used in the regression analysis in Minitab.

3. CAPM for TGT

$$\text{Expected Return} = \text{RF} + \text{Beta} \times (\text{Market Risk Premium})$$

Where:

$$\text{RF} = 0.002$$

$$\text{Market Risk Premium} = 0.005$$

$$\text{Expected Return} = 0.002 + (1.012 \times 0.005) = 0.00706$$

Table: 3

Alpha	Beta	Expected Return	Risk
0.00168	1.012	0.00706	0.0899

4. Interpretation of TGT Alpha and Beta

The beta of TGT is 1.012 and is statistically significant ($p = 0.000$). This means TGT moves almost one-for-one with the market. A 1% increase in market excess return leads to approximately a 1.012% increase in TGT's excess return. Since beta is slightly above 1, TGT has slightly higher systematic risk than the overall market.

This is consistent with the company's business model. As a large retailer operating in a highly competitive and economically sensitive industry, Target's revenues depend heavily on consumer spending and overall economic conditions. Because retail sales fluctuate with economic cycles, the company's stock performance is naturally affected by market movements.

The alpha is 0.00168 but is not statistically significant ($p = 0.823$). This means there is no evidence that TGT earns abnormal returns beyond what is explained by market risk. The stock's return is primarily driven by systematic market risk.

5. CAPM for WMT

Table: 4

Alpha	Beta	Expected Return	Risk
0.00683	0.428	0.00414	0.0533

Interpretation of WMT Alpha and Beta

The beta of WMT is 0.428 and is statistically significant ($p = 0.000$). This means WMT is less sensitive to market movements. A 1% increase in market excess return results in only a 0.428% increase in WMT's excess return. Since beta is well below 1, WMT is considered a defensive stock with lower systematic risk than the market.

This observation is consistent with Walmart's business strategy because it primarily sells basic items such as food and household essentials, unlike Target. As such, it is less affected by market trends because demand for such items will remain steady even in a recession.

The alpha is 0.00683 but is not statistically significant ($p = 0.162$). Therefore, there is no statistical evidence that Walmart earns abnormal returns beyond what is explained by market risk.

6. Optimal Risky Portfolio for CAPM

Table: 5

Weight TGT	Weight WMT	Portfolio Return	Portfolio Risk	Portfolio Sharpe
0.594	0.406	0.005844	0.0657	0.059

An optimum risky portfolio refers to the right combination of risky assets to arrive at the highest Sharpe ratio that is possible. This is to maximise returns per unit of risk.

- The Sharpe ratio of the portfolio constructed using historical average returns (0.1937) is substantially higher than that of the CAPM-based portfolio (0.059). This difference arises primarily because historical average returns are noisy estimates of true expected returns and are highly sensitive to the specific sample period. Even temporary positive shocks or unusually strong performance during the sample can inflate the sample mean, leading to overstated expected returns.

In contrast, CAPM-implied expected returns are determined by systematic market risk and are based on equilibrium asset pricing theory. CAPM assumes that investors are compensated only for exposure to market risk, not for firm-specific or temporary deviations in performance. As a result, CAPM produces more conservative and risk-adjusted expected returns.

8.

Table: 6 (Fama French regression for TGT)

Alpha	Beta	SMB beta	HML beta	Expected Return	Risk
0.00306	1.003	0.058	-0.049	0.007068	0.0899

Expected Return = Alpha + Beta × (Market Risk Premium) + Beta1 x (SMB) + Beta2 X (HML)

Where, Risk free Rate: 0.02%, Risk Premium: 0.5%, SMB: 0.2%, HML: 0.1%

Expected Return = 0.002 + (1.003 × 0.005) + (0.058 x 0.002) + (-0.049 x 0.001) = 0.007082

Table: 7 (Fama French regression for WMT)

Alpha	Beta	SMB beta	HML beta	Expected Return	Risk
0.00481	0.527	0.533	-0.180	0.003389	0.0533

Expected Return = Alpha + Beta × (Market Risk Premium) + Beta1 x (SMB) + Beta2 X (HML)

Where, Risk free Rate: 0.02%, Risk Premium: 0.5%, SMB: 0.2%, HML: 0.1%

Expected Return = 0.002 + (0.527 × 0.005) + (0.533 x 0.002) + (-0.180 x 0.001) = 0.003389

9. Interpretation of TGT:

- The Fama-French regression results indicate that TGT's returns are primarily driven by overall market movements. The market beta of 1.004 is statistically significant ($p < 0.001$) and suggests that TGT moves almost one-for-one with the market. This is consistent with Target's position as a large, mature retailer whose performance is closely tied to general economic conditions and consumer spending trends.
- The estimated alpha is positive but statistically insignificant ($p = 0.689$). This indicates that TGT does not generate abnormal returns beyond what is explained

by exposure to systematic risk factors. In other words, its performance is well explained by market risk rather than firm-specific outperformance.

- The SMB coefficient is small and statistically insignificant ($p = 0.860$), implying that TGT does not exhibit meaningful exposure to the size factor. This aligns with Target's status as a large-cap firm, where size-related risk premia are not expected to be significant.
- Similarly, the HML coefficient is negative but insignificant ($p = 0.786$), suggesting no meaningful tilt toward either value or growth characteristics. The slight negative sign may indicate a mild growth orientation, but the lack of statistical significance means this effect is economically negligible.

10. Optimal Risky Portfolio for French Fama

Table: 8

Weight TGT	Weight WMT	Portfolio return	Portfolio risk	Portfolio Sharpe
0.9413	0.0587	0.006865	0.085981	0.0566

11. Although the historic-average portfolio produces the highest Sharpe ratio (0.1937), this result likely reflects estimation error in historical mean returns, which are unstable and prone to overfitting. In contrast, the CAPM and Fama-French portfolios use risk-adjusted expected returns grounded in asset pricing theory and therefore provide more economically realistic estimates. Between the two model-based portfolios, the CAPM portfolio is preferable as it offers a slightly higher Sharpe ratio (0.059) with lower risk than the Fama-French portfolio. Therefore, a rational investor would likely select the CAPM-based optimal risky portfolio.

SECTION 3

1. The dataset provided already contains the periodic returns of the ETF in the column labelled 'XRT'. These values represent the percentage change in the ETF's price for each time period and were used directly in the subsequent analysis.
2. To account for the time value of money, the excess returns of the ETF were computed using the Fama-French risk-free rate (RF) provided in the dataset. Excess return refers to the return earned by an asset above the risk-free rate of return. For each time period, the excess return of XRT was calculated by subtracting the risk-free rate from the ETF return. This was implemented in Minitab using the Calculator function by generating a new variable defined as **XRT_Excess = XRT - RF**.
- 3.

Measure	Value
Alpha	0.00669
Beta	0.6273
Systematic Variance	0.0026122
Idiosyncratic Variance	0.005486
Total Variance	0.0080808
R ²	0.3232
Information Ratio	0.0904663

4. **The beta estimated using the CAPM model for Target is 1.012**, whereas the beta estimated using the **Index Model is 0.6273**. The CAPM beta is higher because it measures Target's sensitivity to movements in the overall market, while the Index Model beta captures sensitivity to movements in the retail industry through XRT. Since Target's performance may not move one-to-one with the entire market, the industry-specific beta is lower. However, the **R² value of the Index Model (0.3232)** is higher than that of the **CAPM model (0.2856)**. This indicates that the retail ETF explains a greater proportion of variation in Target's excess returns compared to the broader market index. As Target operates within the retail sector, industry-level movements are more relevant in explaining its return variability than overall market movements.

Model	Beta	R ²
CAPM (TGT vs Mkt-RF)	1.012	0.2856
Index Model (TGT vs XRT)	0.6273	0.3232

5. Beta measures the sensitivity of a stock's returns to movements in the chosen index, whereas R² measures the proportion of total variation in the stock's returns explained by the model. A high beta does not necessarily imply a high R², as a stock may be highly sensitive to index movements while still being influenced by significant firm-specific factors. Similarly, a statistically significant beta does not guarantee a high R² value, as significance only indicates the existence of a relationship and not the extent to which the model explains return variability.